

NLP-Based Code Summarization Tool

14-Week Execution Plan & Weekly Roadmap

Academic Year 2024–2025

Phase	Weeks	Focus
Phase 1	Week 1–2	Problem Understanding & Foundations
Phase 2	Week 3–4	Requirement Analysis & System Design
Phase 3	Week 5–6	AST Extraction & Feature Engineering
Phase 4	Week 7–9	NLP Model Integration
Phase 5	Week 10–12	System Integration & Evaluation
Phase 6	Week 13–14	Finalization & Reporting

Technology: Python, Streamlit, HuggingFace Transformers, PyTorch

AI Model: Salesforce CodeT5 (codet5-base-multi-sum)

Domain: Natural Language Processing + Compiler Design

PHASE 1 — Problem Understanding & Foundations

WEEK 1: Problem Definition & Context Understanding**Phase 1**

Activities	Deliverables
Study the problem of manual code documentation	Problem Definition Document
Understand Compiler Design concepts: Lexing, Parsing, AST generation	Initial Project Scope & Objectives
Define project scope: summarizing Python functions using NLP	Notes on AST concepts
Research AST-based code analysis using Python ast library	—
Understand Code Summarization importance	—

WEEK 2: Literature Survey & Technology Selection**Phase 1**

Activities	Deliverables
Study CodeBERT, GraphCodeBERT, CodeT5	Literature Review Table
Research HuggingFace Transformers	Technology Stack Decision
Understand Transfer Learning	Research Gap Statement
Select CodeT5 model	—
Study CodeSearchNet dataset	—
Understand Encoder-Decoder architecture	—

PHASE 2 — Requirement Analysis & System Design

WEEK 3: Requirements & Project Planning**Phase 2**

Activities	Deliverables
Identify functional requirements	SRS Document
Define non-functional requirements	Functional Requirement List
Plan CLI and Web App	requirements.txt
Define JSON storage	—
List required libraries	—

WEEK 4: Architecture & System Design**Phase 2**

Activities	Deliverables
Design pipeline: Input → AST → Preprocess → CodeT5 → Summary	Architecture Diagram
Plan folder structure	Folder Structure Plan
Design Streamlit UI	UI Layout Sketch
Plan module data flow	—
JSON storage format design	—

PHASE 3 — AST Extraction & Feature Engineering

WEEK 5: AST Generation & Parsing Module**Phase 3**

Activities	Deliverables
Implement <code>parser.py</code> using <code>ast</code> library	Working <code>parser.py</code>
Use <code>ast.parse()</code> for validation	Validation results
Handle <code>SyntaxError</code> cases	AST sample outputs
Test with sample functions	—

WEEK 6: AST Serialization & Code Preprocessing**Phase 3**

Activities	Deliverables
Implement <code>ast_serializer.py</code>	Working serializer module
Extract functions/classes/imports	Sample outputs
Implement <code>preprocess_code()</code>	Preprocessing results
Remove docstrings	—
Flatten multiline code	—

PHASE 4 — NLP Model Integration

WEEK 7: Model Setup & Tokenizer Integration**Phase 4**

Activities	Deliverables
Install transformers, torch, streamlit	Working model.py
Load tokenizer	Successful tokenizer load
Load model	Cached model verification

WEEK 8: Summary Generation & Beam Search**Phase 4**

Activities	Deliverables
Implement <code>generate_summary()</code>	Working summary generation
Tokenize code input	Tested summaries
Configure beam search	Verified output quality

WEEK 9: Model Testing & Output Refinement**Phase 4**

Activities	Deliverables
Test 10+ Python functions	Tested samples
Fix model issues	Bug fix log
Complex code testing	Quality output verification

PHASE 5 — System Integration & Evaluation

WEEK 10: CLI Development & Data Storage**Phase 5**

Activities	Deliverables
Build main.py CLI Save samples to JSON Terminal output testing	Working CLI tool samples.json CLI screenshots

WEEK 11: Streamlit Web Application**Phase 5**

Activities	Deliverables
Build app.py Two-column layout Generate Summary button	Working Streamlit app UI screenshot History section

WEEK 12: Integration Testing & Bug Fixing**Phase 5**

Activities	Deliverables
Run complete tests Edge case handling	Bug-free system Final verification

PHASE 6 — Finalization & Reporting

WEEK 13: Final Refinement & Documentation**Phase 6**

Activities	Deliverables
Cleanup code files Write README Prepare demo examples	requirements.txt Fully documented code Demo-ready files

WEEK 14: Final Report & Evaluation Preparation**Phase 6**

Activities	Deliverables
Prepare final report Create slides Practice viva explanation	PDF report Presentation slides Demo ready